

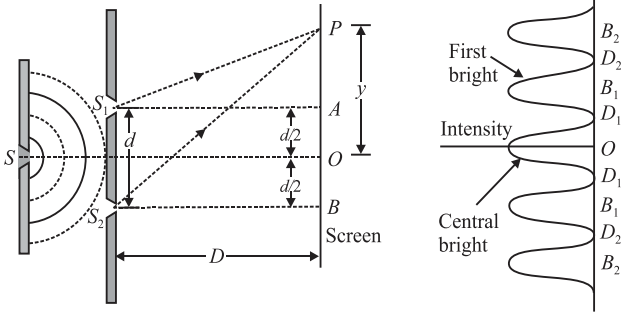


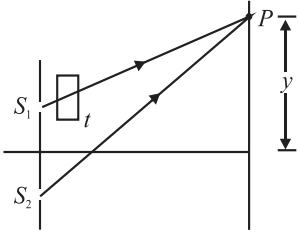
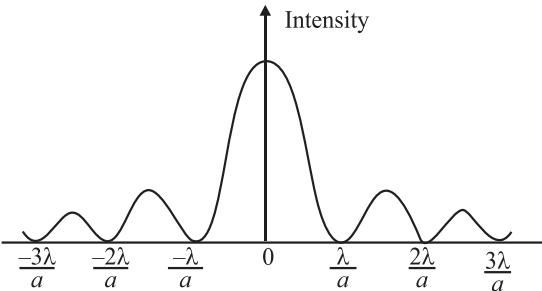
CHAPTER

10

Wave optics

Sr. No.	Concept	Formulas	Other information										
1.	Optical path and wavelength	Wavelength in medium, $\lambda' = \frac{\lambda}{\mu}$ Optical path (in vacuum) = $\mu \times$ Path in medium	λ = wavelength of the light in vacuum. μ = refractive index of medium.										
2.	Amplitude and Intensity due to superposition of waves	Resultant amplitude: $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$ Resultant intensity: $I \propto A^2$ $I = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2} \cos \phi$ When, $I_1 = I_2 = I_0, I = 2I_0 (1 + \cos \phi) = 4I_0 \cos^2 \frac{\phi}{2}$	A_1, A_2 = amplitude of waves. ϕ = phase difference between the waves. I_1, I_2 = intensity of waves.										
3.	Constructive and destructive interference of light waves	<table><tr><th>Constructive Interference</th><th>Destructive Interference</th></tr><tr><td>$\Delta x = 0, \lambda, 2\lambda, 3\lambda, \dots, n\lambda$ $n = 0, 1, 2, 3, \dots$</td><td>$\Delta x = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots, \frac{(2n-1)\lambda}{2}$ $n = 1, 2, 3, \dots$</td></tr><tr><td>$\Delta \phi = 0, 2\pi, 4\pi, \dots, 2n\pi$</td><td>$\Delta \phi = 0, 3\pi, 5\pi, \dots, (2n-1)\pi$</td></tr><tr><td>$A_{\max} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos 0}$ $A_{\max} = A_1 + A_2$</td><td>$A_{\min} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \pi}$ $A_{\min} = A_1 - A_2$</td></tr><tr><td>$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$</td><td>$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$</td></tr></table>	Constructive Interference	Destructive Interference	$\Delta x = 0, \lambda, 2\lambda, 3\lambda, \dots, n\lambda$ $n = 0, 1, 2, 3, \dots$	$\Delta x = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots, \frac{(2n-1)\lambda}{2}$ $n = 1, 2, 3, \dots$	$\Delta \phi = 0, 2\pi, 4\pi, \dots, 2n\pi$	$\Delta \phi = 0, 3\pi, 5\pi, \dots, (2n-1)\pi$	$A_{\max} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos 0}$ $A_{\max} = A_1 + A_2$	$A_{\min} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \pi}$ $A_{\min} = A_1 - A_2$	$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$	$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$	Δx = path difference $\Delta \phi$ = phase difference $A_{\max}$ = maximum amplitude $A_{\min}$ = minimum amplitude $I_{\max}$ = maximum intensity $I_{\min}$ = minimum intensity
Constructive Interference	Destructive Interference												
$\Delta x = 0, \lambda, 2\lambda, 3\lambda, \dots, n\lambda$ $n = 0, 1, 2, 3, \dots$	$\Delta x = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots, \frac{(2n-1)\lambda}{2}$ $n = 1, 2, 3, \dots$												
$\Delta \phi = 0, 2\pi, 4\pi, \dots, 2n\pi$	$\Delta \phi = 0, 3\pi, 5\pi, \dots, (2n-1)\pi$												
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4.	Young's Double Slit Experiment	 For a bright fringe, path difference; $\Delta x = n\lambda$ For a dark fringe; $\Delta x = (2n-1)\frac{\lambda}{2}, n = 1, 2, 3, \dots$ Distance of n^{th} bright fringe; $y_n = n \frac{D\lambda}{d}, n = 1, 2, 3, \dots$ Distance of n^{th} dark fringe; $y'_n = (2n-1) \frac{D\lambda}{2d}, n = 1, 2, 3, \dots$ Fringe width; $\beta = \frac{D\lambda}{d}$ If whole apparatus is immersed in liquid of refractive index n; $\lambda' = \frac{\lambda}{n}$ New fringe width; $\beta' = \frac{\beta}{n}$ Angular position of n^{th} bright fringe; $\theta_n = \frac{y_n}{D} = \frac{n\lambda}{d}$ Angular position of n^{th} dark fringe; $\theta'_n = \frac{y'_n}{D} = (2n-1) \frac{\lambda}{2d}$ Angular width of a fringe = $\frac{\beta}{D} = \frac{\lambda}{d}$	D = distance between screen and slit d = separation between two slits. y = position of superimposition of waves p = path difference λ = wavelength of the light source. β = fringe width
5.	Relation between intensity and amplitude of waves	Intensity $\propto$ width of slit $\propto$ (amplitude)2 $\Rightarrow I \propto W \propto A^2 \Rightarrow \frac{I_1}{I_2} = \frac{W_1}{W_2} = \frac{A_1^2}{A_2^2}$ $\frac{I_{\max}}{I_{\min}} = \left[\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right]^2 = \left[\frac{A_1 + A_2}{A_1 - A_2} \right]^2 = \left[\frac{A_{\max}}{A_{\min}} \right]^2$ Fringe visibility = $\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$	A_1, A_2 = amplitude of waves. I_1, I_2 = intensity of waves. W = width of the slit

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6.	Displacement of interference fringes when a slab is introduced in path of one of the interfering waves	 The extra path difference due to slab: $\Delta x = \mu t - t = (\mu - 1)t$ Displacement of the central bright fringe, $\Delta y = \frac{D}{d}(\mu - 1)t$	β = fringe width λ = wavelength t = thickness of the slab μ = refractive index of the slab.
7.	Diffraction due to a single slit	(i) Condition for n^{th} minimum is; $a \sin \theta = n\lambda$ where $n = 1, 2, 3, \dots$ (ii) Angular position of n^{th} minimum; $\theta_n = \frac{n\lambda}{a}$ (iii) Distance of n^{th} minimum from the centre of the screen, $x_n = \frac{nD\lambda}{a}$ (iv) Angular position of n^{th} secondary maximum, $\theta'_n = (2n + 1) \frac{\lambda}{2a}$ (v) Distance of n^{th} secondary maximum from the centre of the screen, $x'_n = (2n + 1) \frac{D\lambda}{2a}$ (vi) Width of central maximum, $\beta_0 = 2\beta = \frac{2D\lambda}{a}$ (vii) Angular spread of central maximum on either side, $\theta = \pm \frac{\lambda}{a}$ (viii) Total angular spread of central maximum, $2\theta = \frac{2\lambda}{a}$ 	a = slit width D = distance between screen and slit d = separation between two slits. β = fringe width. θ = diffraction angle λ = wavelength

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8.	Brewster's law	According to brewster's law: $\mu = \tan \theta_p \Rightarrow \theta_p = \tan^{-1} \mu$	θ_p = polarization or Brewster's angle μ = refractive index of refracting medium
9.	Malus Law	Law of Malus, $I = I_0 \cos^2 \theta$	I_0 = incident intensity of plane polarized light. I = intensity of transmitted light through analyser θ = angle between the axes of polarizer and analyser

